Dataset Details

Uplift: Student Success Assistant

Open University Learning Analytics Dataset (OULAD). Counts in this document are read from the data files and the build metrics by `scripts/build_docs.py`.

1. Dataset name

Open University Learning Analytics Dataset (OULAD), the dataset named in the problem statement.

2. Dataset source

The Open University (United Kingdom), Knowledge Media Institute. Published by Kuzilek, Hlosta and Zdrahal (2017), *Open University Learning Analytics dataset*, Scientific Data 4:170171, doi:10.1038/sdata.2017.171. Licence: Creative Commons Attribution 4.0 (CC BY 4.0). The data is anonymised by the publisher; student IDs are not linked to real people.

3. Dataset link

- Primary: https://analyse.kmi.open.ac.uk/open_dataset (also mirrored on Figshare, as given in the brief)
- Alternative: the OULAD GitHub mirror referenced in the brief
- Paper: https://doi.org/10.1038/sdata.2017.171

4. Data format

Seven CSV files (comma separated, header row, UTF-8), relational and joined on `code_module`, `code_presentation`, `id_student`, `id_assessment` and `id_site`. Place them in `data/raw/` (not committed; see the README).

File	What one row is	Rows (full dataset)	Rows (`data/sample_data`)
`courses.csv`	a module presentation and its length in days	22	22
`assessments.csv`	an assessment of a presentation (type, deadline day, weight)	206	206
`vle.csv`	a learning material in the virtual learning environment	6,364	6,364
`studentInfo.csv`	a student's enrolment on a presentation: demographics and final result	32,593	880

File	What one row is	Rows (full dataset)	Rows (`data/sample_data`)
`studentRegistration.csv`	registration and (if any) unregistration day for an enrolment	32,593	880
`studentAssessment.csv`	a submission: day submitted and score	173,912	4,644
`studentVle.csv`	a student's clicks on one material on one day	10,655,280	294,849

Scope: 4 presentations (2013B, 2013J, 2014B, 2014J) across 7 modules (AAA to GGG), 32,593 enrolments in total.

5. Key fields

Table	Fields	Use in this project
studentInfo	`id_student, code_module, code_presentation, gender, region, highest_education, imd_band, age_band, num_of_prev_attempts, studied_credits, disability, final_result`	`final_result` is the **label** (Distinction / Pass / Fail / Withdrawn). Previous attempts and credits are features. The six demographic fields are **sensitive**: never model inputs, never sent to the LLM; used only for the fairness audit and as retrieval filters.
studentRegistration	`code_module, code_presentation, id_student, date_registration, date_unregistration`	Registration timing is a feature. `date_unregistration` only decides who is **still enrolled** at a checkpoint (scope); it is never a feature.
assessments	`code_module, code_presentation, id_assessment, assessment_type, date, weight`	Deadlines (TMA, CMA, Exam) and weights: which work was due by the checkpoint.
studentAssessment	`id_assessment, id_student, date_submitted, is_banked, score`	Scores, missed work, lateness, score trend, only for submissions on or before the checkpoint day.
vle	`id_site, code_module, code_presentation, activity_type, week_from, week_to`	Material types (content, resource, forum, quiz...) and planned weeks.
studentVle	`code_module, code_presentation, id_student, id_site, date, sum_click`	Daily clicks: weekly engagement, inactivity streaks, trends, material coverage, up to the checkpoint.
courses	`code_module, code_presentation, module_presentation_length`	Course length: checkpoints are fractions of it.

6. Dataset purpose

OULAD links, for each enrolment, the student profile, the assessment record, day-by-day online learning activity and the final result. That is what the brief needs:

- **Early warning**: predict withdrawal and the final outcome from what is known part-way through a course.
- **Similar-case retrieval (RAG)**: every finished enrolment is a historical case with a known outcome, so "what happened to students like this one" can be answered from real records.
- **Prioritisation, explanation and trends**: assessment and engagement signals rank and explain risk; weekly activity shows when struggling students diverge.
- **Fairness**: the demographic fields let us audit the model's errors by group, without using them as inputs.

7. Data preprocessing performed

Implemented in `src/pipeline/pipeline.py` (`python -m src.cli prepare`), once per checkpoint.

7.1 Checkpoints ("as of" snapshots). A snapshot is taken at 4 points of each presentation: 15% (week 5), 25% (week 9), 40% (week 15), 60% (week 22)of its length. Every feature uses only rows dated on or before the snapshot day.

7.2 Population scoping (the most important step). Students who had already unregistered by the snapshot day cannot be helped by an early warning. They are kept in the data, marked `in_scope = 0`, and excluded from modelling, evaluation, retrieval and ranking. At week 15: 32,593 enrolments, of which 25,204 are in scope; 7,389 had already left (3,097 before the course started).

Checkpoint	Course week	Students still enrolled (in scope)
c15	5	27,157
c25	9	26,241
c40	15	25,204
c60	22	23,926

7.3 Leakage controls (recorded in the build metrics): - assessment and VLE rows filtered to date <= snapshot_day - students unregistered on/before snapshot_day excluded from modelling scope (in_scope=0) - date_unregistration used for scoping only; never a feature - final_result used as label only - peer percentiles computed from in-scope peers at snapshot time

7.4 Cleaning. - Missing is not zero: students with no scored assessment keep empty scores plus a `no_scores_yet` flag. - Banked results (carried over from earlier attempts) are counted separately. - `studentVle.csv` (the large table) is read with compact data types and joined on integer keys, so the full build fits in under 2 GB of memory. - Categorical values are validated against fixed vocabularies (modules, regions, education levels, age bands).

7.5 Feature engineering. 66 numeric features in three groups, plus the module:

- *Enrolment*: previous attempts, credits, how early the student registered, late registration.
- *Academic*: assessments due, submitted, missed (TMA and CMA separately), average / weighted / last / lowest score, score trend, lateness, cohort percentiles.

- *Engagement*: weekly clicks on a dense week grid (zero-activity weeks included), active days, recent vs early activity, momentum, inactivity streaks, days since last activity, activity by material type, share of scheduled materials opened, cohort percentiles.

▶ All features

7.6 Labels. Withdrawal after the snapshot (binary, among students still enrolled) and the four-class final outcome. "Poor outcome" (fail or withdraw) is used for prioritisation.

7.7 Retrieval documents. Each finished, in-scope enrolment becomes one case: its snapshot features, discretised descriptor tokens (for example `missed_most`, `engagement_declining`) and a short narrative for the dense channel. Numbers never reach the text channels; they are banded relative to the cohort. See the Design Document, section 4.

7.8 Temporal split. Train on 2013B + 2013J, tune on 2014B, test once on 2014J (the latest presentation, never used for any choice). Historical retrieval only searches presentations earlier than the query.

8. Sample data (`data/sample_data/`)

A real slice of OULAD in the original seven-table format, committed so the pipeline can be run without downloading the full dataset: a fixed number of students drawn (seed 42) from every module presentation, with all their rows; the reference tables are complete. Made by `scripts/make_sample_data.py`. Run the whole pipeline on it with:

```
SSA_HOME=.sample SSA_RAW_DIR=data/sample_data python -m src.cli build --no-dense
```

It is far too small for meaningful metrics; every result reported in this submission comes from the full dataset.